SAFEGUARDING GUIDING

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1. Introduction

Over the past two decades Sequential Monte Carlo (SMC) methods have gained increasing popularity. In this paper, we are specifically interested in state-space models using the *guided particle filter*. A state-space model consists of a latent Markovian process $\{X_t\}$ and an observation process $\{Y_t\}$ $(t = 0, 1, 2, \ldots)$, where the distribution of Y_t depends on X_t. Algorithm 10.5 in [CP20] (restated here as Algorithm 1.3) provides the steps of the guided particle filter to approximate the filtering distribution, i.e. the distribution of $X_t \mid Y_0, \ldots, Y_t$. The algorithm relies on choosing mutation kernels M_t and setting potentials G_t equal to

$$G_t(x_{t-1}, x_t) = f_t(y_t \mid x_t) \frac{P_t(x_{t-1},\, \mathrm{d}x_t)}{M_t(x_{t-1},\, \mathrm{d}x_t)},$$

where f_t is the emission density at time t (which we assume to exist) and P_t is the Markov transition kernel for the latent process $\{X_t\}$. Implicitly, we assume M_t to be chosen such that the Radon-Nikodym derivative in the preceding display is well defined. The output of the particle filter readily yields an unbiased estimator of the likelihood, which we will denote by $\hat{L}$.

The Bootstrap filter corresponds to the special case where M_t is chosen to equal P_t. However, choosing the kernel M_t to take the future observation y_t into account may lead to a more efficient algorithm, for example quantified by an increased effective sample size (ESS). *Guiding*, also known as *twisting*, provides a generic framework to do so. The basic idea consists of choosing a *guiding function* g_t and take M_t equal to P_t^g, where

$$P_t^{g_t}(x_{t-1},\, \mathrm{d}x_t) = \frac{P_t(x_{t-1},\, \mathrm{d}x_t) g_t(x_t)}{\int P_t(x_{t-1},\, \mathrm{d}x_t) g_t(x_t)}$$

assuming g_t to be such that the denominator is finite. The Markovian process with these transition kernels will be referred to as the *guiding process* induced by g_t. Doob's h-transform consists of choosing the guiding function such that the guided process is in fact the exact process conditioned on the observation y_t; the corresponding guiding function will be denoted by h_t. It is well known that h_t is only known in closed form in a few special cases. In applications we therefore look for a "good" guiding function. For sampling efficiency, it is advantageous to choose g_t such that sampling from $P_t^{g_t}$ is feasible. It may however be the case that intuitively reasonable choices that facilitate easy sampling result in the unbiased estimator $\hat{L}$ having infinite variance. In Section 2 we give an example.

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In this paper we study safe-guarding the guided particle filter by replacing g_t with $g_t + \epsilon$ for a (small) value of ϵ. This idea has been briefly stated in some works on SMC ([GJL17], [LHV18]), without much detailed study. In our first part of this paper we consider discrete-time state space models and introduce an algorithm for choosing ϵ adaptively. We provide theoretical result to assess whether inclusion of a strictly positive ϵ leads to a decrease in χ^2 divergence between P^h and $P^{g+\epsilon}$ and therefore an increase in ESS. We introduce an algorithm for adaptive tuning of ϵ and illustrate this using numerical examples.

In the second part of the paper we extend our results to the setting where the latent process is a continuous time process defined as solution to the stochastic differential equation (SDE)

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t.$$

A guiding function induces a change of measure from the forward process, denoted by P, to the measure P^g of a process called the *guided* process. Following work on exponential change of measure for continuous time stochastic processes ([PR02], [vdMS25]), under P^g the process X satisfies the SDE

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + qq^\top r(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t^\circ,$$

with $r(t,x) = \nabla_x \log g(t,x)$ and W° a P^g-Wiener process. Similar to the discrete-time case we study the effect of changing g to $g + \epsilon$ and give an algorithm for adaptively tuning ϵ.

1.1. Contributions.

1.2. Outline.

1.3. Frequently used notation. $\varphi(x; \mu, \Sigma)$ denotes the density of the $\mathcal{N}(\mu, \Sigma)$-distribution, evaluated at x.

For a Markov kernel P and compatible function g we write $(Pg)(x) = \int P(x, \mathrm{d}y)g(y)$.

2. Stochastic volatility example

As illustrative example to motivate our work, we consider the well-known stochastic volatility model. For $t = 0, 1, \ldots$ define

$$\begin{aligned} R_t &= e^{X_t/2} U_t \\ X_t &= \omega + \psi X_{t-1} + \eta W_t \end{aligned}, \tag{1}$$

where $\{U_t\}$ and $\{W_t\}$ are independent sequences of independent $\mathcal{N}(0,1)$-distributed random variables. The process $\{X_t\}$ is latent, while $\{R_t\}$ is observed. The observation equation is equivalent to its log-transformed version given by

$$V_i := \log\left(R_i^2\right) = X_i + U_i', \quad 1 \le i \le n, \tag{2}$$

where $U_i' := \log\left(U_i^2\right)$, $i = 1, \ldots, n$.

Suppose for simplicity that x_0 is fixed and we wish to forward propagate a particle at x_0 in view of the observation v_1 at time 1. Let $P(x_0, \mathrm{d}x_1)$ be the Markov kernel corresponding to the conditional distribution

$$X_1 \mid X_0 = x_0 \;\sim\; \mathcal{N}(\omega + \psi x_0,\ \eta^2).$$

Algorithm 1 Guided particle filter (GPF)

Operations involving index n must be performed for $n = 1, \ldots, N$.

$X_0^n \sim \mathbb{M}_0(\mathrm{d}x_0)$

$w_0^n \leftarrow G_0(X_0^n)$ $\triangleright$ where $G_0(x_0) := \dfrac{\mathbb{P}_0(\mathrm{d}x_0)\, f_0(y_0 \mid x_0)}{\mathbb{M}_0(\mathrm{d}x_0)}$

$W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$

for $t = 1$ **to** T **do**

  if $\mathrm{ESS}(W_{t-1}^{1:N}) < \mathrm{ESS}_{\min}$ **then**

    $A_t^{1:N} \leftarrow \mathrm{resample}(W_{t-1}^{1:N})$

    $\widehat{w}_{t-1}^n \leftarrow 1$

  else

    $A_{t-1}^n \leftarrow n$

    $\widehat{w}_{t-1}^n \leftarrow w_{t-1}^n$

  end if

  $X_t^n \sim M_t(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$

  $w_t^n \leftarrow \widehat{w}_{t-1}^n\, G_t(X_{t-1}^{A_t^n}, X_t^n)$ $\triangleright$ where

$$G_t(x_{t-1}, x_t) := \frac{f_t(y_t \mid x_t)\, P_t(x_{t-1}, \mathrm{d}x_t)}{M_t(x_{t-1}, \mathrm{d}x_t)}$$

  $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$

end for

The distribution of X_1, conditioned on v_1 and x_0 is given by the twisted kernel

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1)\, h(x_1)}{\int P(x_0, dx_1)\, h(x_1)},$$

where

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v_1, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

It is easily verified that

$$h(x_1) = f_{\chi^2(1)}\big(e^{v_1 - x_1}\big)\, e^{v_1 - x_1}, \tag{3}$$

with $f_{\chi^2(1)}$ denoting the density of the $\chi^2(1)$-distribution. As sampling from P^h is non-trivial, we use a Gaussian approximation to the distribution of $\log Z^2$ to obtain the guiding function g defined by

$$g(x_1) = \varphi(v_1;\ \mu + x_1,\ \sigma^2),$$

where $\mu := \mathbb{E} \log Z^2 \approx -1.27$ and $\sigma^2 := \mathrm{Var}\big(\log Z^2\big) = \pi^2/2$. This induces the twisted measure

$$P^g(x_0,\, \mathrm{d}x_1) = \frac{P(x_0,\, \mathrm{d}x_1)\, g(x_1)}{\int P(x_0,\, \mathrm{d}x_1)\, g(x_1)}.$$

Contrary to P^h, sampling from P^g is easy. A simple calculation reveals that

$$P^g(x_0,\, \mathrm{d}x_1) \sim \mathcal{N}\left(\mu_g, \sigma_g^2\right),$$

with

$$\sigma_g^2 = \frac{\eta^2 \sigma^2}{\eta^2 + \sigma^2} \qquad \text{and} \qquad \mu_g = \frac{\sigma^2(\omega + \psi x_0) + \eta^2(v - \mu)}{\eta^2 + \sigma^2}.$$

earlier denoted by $\mu^\star$ and $\sigma^\star$ look up where this has been proposed in the literature and cite this

Suppose $X_1^1, \ldots, X_1^N$ are sampled independently from $P^g(x_0, \mathrm{d}x_1)$. An unbiased estimator for the likelihood is given by

$$\hat{L} := \frac{1}{N} \sum_{n=1}^{N} G(x_0, X_1^n),$$

where

$$G(x_0, x_1) = h(x_1) \frac{P(x_0, \mathrm{d}x_1)}{P^g(x_0, \mathrm{d}x_1)} = h(x_1) \frac{(Pg)(x_0)}{g(x_1)}.$$

Define

$$m(0) = \mathbb{E}_{P^g} [G(x_0, X_1)]^2 , \tag{4}$$

the notation being chosen such that it matches with more general notation introduced later.

Proposition 1.

$$m(0) < \infty \iff \eta < \sigma.$$

Proof. double check proof We have

$$\begin{aligned} m(0) &= [(Pg)(x_0)]^2 \int \left(\frac{h(x_1)}{g(x_1)} \right)^2 P^g(x_0, \mathrm{d}x_1) \\ &= (Pg)(x_0) \int \frac{h(x_1)^2}{g(x_1)} \, P(x_0, \mathrm{d}x_1) \end{aligned}$$

From (3) we get $h(x_1)^2 = \frac{1}{2\pi} e^{-(v_1 - x_1)} e^{-e^{v_1 - x_1}}$. Writing $t = x_1 - v_1$,

$$\frac{h(x_1)^2}{g(x_1)} \varphi(x_1; \omega + \psi x_0, \eta^2) \ \propto \ \exp\left(t - e^{-t} + \frac{(t+\mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2} \right)$$

where the proportionality is with respect to x_1.

We analyse the tails:

- *Right tail* $t \to +\infty$. The term $e^{-t} \to 0$, and expanding the quadratics the dominant behaviour is

$$\exp\left(t^2 \left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} \right) + O(t) \right).$$

  This is integrable if and only if the coefficient of t^2 is strictly negative, i.e.

$$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

- *Left tail* $t \to -\infty$. The term $e^{-t} \to +\infty$ dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

□

It follows that for $\eta \geq \pi/\sqrt{2} \approx 2.221$ the estimator $\hat{L}$ has infinite variance. One possible solution is to choose σ strictly larger than η. In this particular example we can do all calculations but this is generally not the case.

3. Safe guarding (discrete time case)

The basic idea is to start with a a function g that enables sampling from P^g. Next, we consider guiding using $g+\epsilon$ for $\epsilon>0$.

Proposition 2. $P^{g+\epsilon}$ is a mixture of P^g and P:

$$P^{g+\varepsilon}(x_0, dx_1) = \lambda_g(\varepsilon) P^g(x_0, dx_1) + (1-\lambda_g(\varepsilon)) P(x_0, dx_1),$$

with $\lambda_g(\varepsilon, x_0) = c/(c+\varepsilon)$ and $c = (Pg)(x_0)$.

Writing c rather than $(Pg)(x_0)$ reduces notation in what follows.

Proof. This follows from

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{P(x_0,\, \mathrm{d}x_1)(g(x_1)+\epsilon)}{\int P(x_0,\, \mathrm{d}x_1)(g(x_1)+\epsilon)} = \frac{P(x_0,\, \mathrm{d}x_1) g(x_1)}{c+\epsilon} + \frac{P(x_0,\, \mathrm{d}x_1)}{c+\epsilon}.$$

□

Similar to the derivations in the preceding section, we define the potential

$$\begin{aligned} G_\varepsilon(x_0, x_1) &:= h(x_1)\, \frac{P(x_0,\, \mathrm{d}x_1)}{P^{g+\varepsilon}(x_0,\, \mathrm{d}x_1)} \\ &= h(x_1)\, \frac{\int P(x_0,\, \mathrm{d}x_1)\, g(x_1)\, \mathrm{d}x_1 + \varepsilon}{g(x_1)+\varepsilon} = h(x_1)\, \frac{c+\varepsilon}{g(x_1)+\varepsilon}. \end{aligned} \tag{5}$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^{N} G_\varepsilon\left(x_0, X_1^{(n)}\right), \qquad \left(X_1^{(n)}\right)_{n=1}^{N} \sim P^{g+\varepsilon}(x_0,\, \mathrm{d}x_1).$$

For *any* $\varepsilon \geq 0$, $\hat{L}$ is an unbiased estimator for the likelihood:

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1)\, \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} P^{g+\varepsilon}(x_0, dx_1) = (Ph)(x_0).$$

Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} \left[G_\varepsilon(x_0, X_1)\right]^2.$$

By expanding we get

$$m(\varepsilon) = \int h(x_1)^2 \left(\frac{c+\varepsilon}{g(x_1)+\varepsilon}\right)^2 \frac{P(x_0,\, \mathrm{d}x_1)\,(g(x_1)+\varepsilon)}{c+\varepsilon} = \big(c+\epsilon\big) I(\epsilon, x_0), \tag{6}$$

where

$$I(\varepsilon, x_0) = \int \frac{h(x_1)^2}{g(x_1)+\varepsilon}\, P(x_0,\, \mathrm{d}x_1).$$

Note that Equation (4) is a special case. Set $\bar{m}(\varepsilon) = \log m(\varepsilon)$.

Example 3. We return to the stochastic volatility example of Section 2. Straightforward computation gives

$$c = \varphi(v; \mu+\omega+\psi x_0; \sigma^2+\eta^2).$$

Moreover, $I(x_0, \epsilon)$ can be numerically approximated using Gauss-Hermite quadrature. In Figure 1 and Figure 2 we plot $\bar{m}$ for $\eta = 1$ and $\eta = 3$ respectively. In the former case we know $\bar{m}(0) < \infty$ while in the latter case $\lim_{\epsilon \downarrow 0} \bar{m}(\epsilon) = \infty$. In both cases $\bar{m}$ has a unique minimiser.

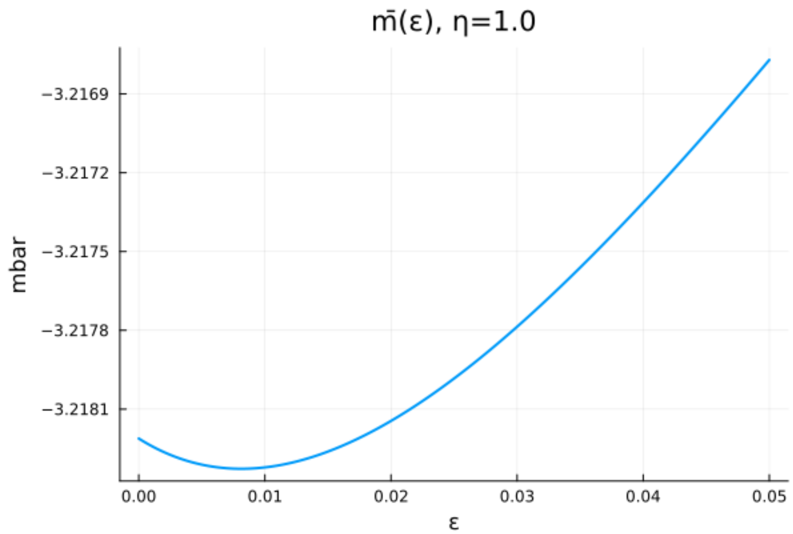


FIGURE 1. Visualisation int he stochastic volatility example of $\epsilon \mapsto \bar{m}$ when $\eta = 1$.

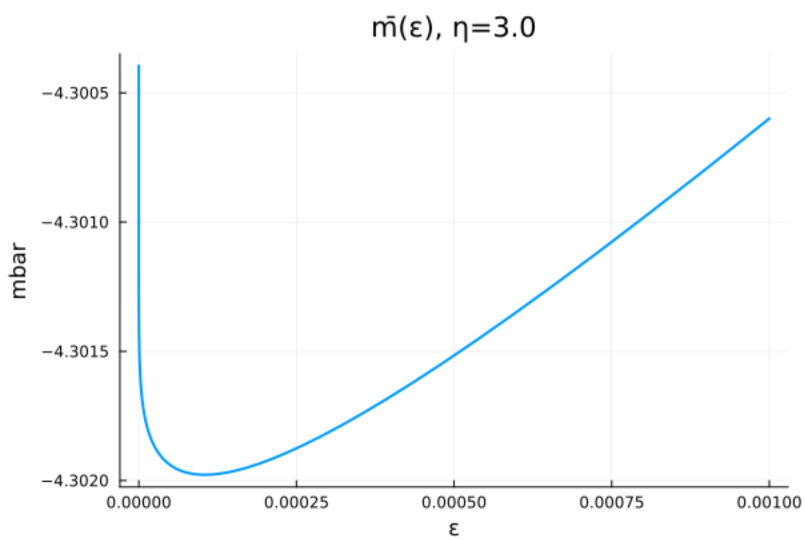


FIGURE 2. Visualisation int he stochastic volatility example of $\epsilon \mapsto \bar{m}$ when $\eta = 3$.

3.1. **Importance sampling quality: KL and χ^2 divergences.** The natural divergences for assessing importance sampling quality of $\hat{L}$ are the KL divergence and the χ^2

divergence between the target P^h and the proposal $P^{g+\varepsilon}$:

$$\mathrm{KL}(P^h \| P^{g+\varepsilon}) = \int \log \frac{\mathrm{d}P^h}{\mathrm{d}P^{g+\varepsilon}}\, \mathrm{d}P^h,$$
$$\chi^2(P^h \| P^{g+\varepsilon}) = \int \left(\frac{\mathrm{d}P^h}{\mathrm{d}P^{g+\varepsilon}}\right)^2 \mathrm{d}P^{g+\varepsilon} - 1.$$

The χ^2-divergence is directly related to the second moment of the potential.

Lemma 4.

$$\chi^2(P^h \| P^{g+\varepsilon}) = \frac{m(\varepsilon)}{[(Ph)(x_0)]^2} - 1,$$

Proof. We have

$$\frac{P^h(x_0, \mathrm{d}x_1}{P^{g+\varepsilon}(x_0, \mathrm{d}x_1)} = \frac{h(x_1)/(Ph)(x_0)}{(g(x_1)+\varepsilon)/((Pg)(x_0)+\varepsilon)} = \frac{G_\varepsilon(x_0, x_1)}{(Ph)(x_0)},$$

where the final equality follows from Equation (5). □

We have add reference

$$ESS \approx \frac{N}{1 + \chi^2(P^h \| P^{g+\varepsilon})}.$$

So minimising $m(\varepsilon)$ at each SMC step is equivalent to maximising ESS.

Using $\log x \le x - 1$ (and thus $2\log x \le x^2 - 1$)

$$2\mathrm{KL}(P^h \| P^{g+\varepsilon}) \le \int \left(\frac{dP^h}{dP^{g+\varepsilon}}\right)^2 dP^h - 1.$$

3.2. **Theoretical Properties of $m(\varepsilon)$.** Throughout this section x_0 is fixed and therefore omitted in some notation. Let $J(\epsilon) = \int h(x)^2/(g(x)+\epsilon)^2\, P(\,\mathrm{d}x)$, so that

$$m(\epsilon) = (c+\epsilon)\, I(\varepsilon), \qquad m'(\epsilon) = I(\epsilon) - (c+\epsilon)\, J(\epsilon).$$

Lemma 5. If $\int h(x_1)^2 P(x_0, \mathrm{d}x_1) < \infty$ then $m(\epsilon) < \infty$ for any $\epsilon > 0$.

Proof.

$$m(\varepsilon) \le \frac{c+\epsilon}{\epsilon} \int h(x_1)^2\, P(x_0, \mathrm{d}x_1) < \infty.$$

□

Proposition 6 (Log-convexity of I)**.** The function $\varepsilon \mapsto I(\varepsilon)$ is strictly decreasing, strictly convex, and log-convex on $[0, \infty)$: for all $\varepsilon_1, \varepsilon_2 \ge 0$,

$$I\left(\frac{\epsilon_1 + \epsilon_2}{2}\right) \le \sqrt{I(\epsilon_1) \cdot I(\epsilon_2)}.$$

Proof. Strictly decreasing: $I'(\epsilon) = -J(\epsilon) < 0$.

Strictly convex: $I''(\epsilon) = 2\int h^2/(g+\epsilon)^3\, \mathrm{d}P > 0$.

Log-convex: By the arithmetic-geometric inequality, $g(x)+\frac{\epsilon_1+\epsilon_2}{2} \geq \sqrt{(g(x)+\epsilon_1)(g(x)+\epsilon_2)}$, so

$$\begin{aligned} I\left(\frac{\varepsilon_1+\varepsilon_2}{2}\right) &\leq \int \frac{h^2}{\sqrt{(g+\varepsilon_1)(g+\epsilon_2)}}\,\mathrm{d}P \\ &= \int \frac{h}{\sqrt{g+\epsilon_1}} \cdot \frac{h}{\sqrt{g+\epsilon_2}}\,\mathrm{d}P \leq \sqrt{I(\epsilon_1)\cdot I(\varepsilon_2)}, \end{aligned}$$

where the last step follows from the Cauchy-Schwarz inequality. □

Proposition 7 (Boundary behaviour)**.** As $\varepsilon \to \infty$, $m(\varepsilon) \to \int h^2\,dP$. The value $m(0)$ may be finite or infinite:

(1) If $h^2/g \in L^1(P)$ then $m(0) = c \cdot I(0) < \infty$.
(2) If $h^2/g \notin L^1(P)$ then $m(0) = \infty$, and in this case $m'(\varepsilon) \to -\infty$ as $\varepsilon \downarrow 0$.

In case 2, $m(\varepsilon) < \infty$ for all $\varepsilon > 0$ by Proposition **??**, so m necessarily has a minimum at some $\varepsilon^* \in (0, \infty)$.

Proof. As $\varepsilon \to \infty$, dominated convergence gives $I(\varepsilon) \to 0$ and $(c+\varepsilon)I(\varepsilon) \to \int h^2\,dP$ by a standard calculation. For case 2, $I(\varepsilon) \uparrow \infty$ as $\varepsilon \downarrow 0$ by monotone convergence, while $(c+\varepsilon)J(\varepsilon)$ stays bounded, so $m'(\varepsilon) \to -\infty$. □

Proposition 8 (Unimodality of m)**.** $m(\varepsilon)$ has at most one critical point $\varepsilon^* \in (0, \infty)$. A sufficient condition is that $\varepsilon \mapsto I(\varepsilon)/J(\varepsilon)$ is strictly increasing, in which case $m'(\varepsilon) = 0$ has at most one solution.

Proof. The critical point equation is $m'(\varepsilon) = 0$, i.e. $I(\varepsilon) = (c+\varepsilon)J(\varepsilon)$, equivalently

$$\frac{I(\varepsilon)}{J(\varepsilon)} = c + \varepsilon. \tag{7}$$

The right-hand side is strictly increasing in ε. If $\varepsilon \mapsto I(\varepsilon)/J(\varepsilon)$ is also strictly increasing, then the left-hand side is strictly increasing and (7) has at most one solution. □

Remark 9. The condition that I/J is strictly increasing is equivalent to $I(\varepsilon)J'(\varepsilon) < I'(\varepsilon)J(\varepsilon)$, i.e. $d\log(I/J)/d\varepsilon > 0$. Substituting:

$$\frac{d}{d\varepsilon}\frac{I(\varepsilon)}{J(\varepsilon)} = \frac{-J(\varepsilon)^2 + I(\varepsilon)\cdot 2\int h^2/(g+\varepsilon)^3\,dP}{J(\varepsilon)^2} = 2\,\frac{I(\varepsilon)\int h^2/(g+\varepsilon)^3\,dP - J(\varepsilon)^2}{J(\varepsilon)^2}.$$

By the Cauchy-Schwarz inequality applied to the measure $\nu(dx) = h^2/(g+\varepsilon)^2\,dP(x)$:

$$J(\varepsilon)^2 = \left(\int \frac{1}{g+\varepsilon}\,d\nu\right)^2 \leq \left(\int \frac{1}{(g+\varepsilon)^2}\,d\nu\right)\cdot \nu(\mathcal{X}) = \int \frac{h^2}{(g+\varepsilon)^3}\,dP \cdot I(\varepsilon),$$

which gives $d(I/J)/d\varepsilon \geq 0$, with equality only when $1/(g+\varepsilon)$ is constant ν-a.s., i.e. when g is constant P-a.s. on the support of h. Hence I/J *is strictly increasing whenever g is not P-a.s. constant on the support of* h, confirming unimodality of $m(\varepsilon)$ under this mild non-degeneracy condition.

Corollary 10. Under the non-degeneracy condition of the remark above, $m(\varepsilon)$ is strictly unimodal: it has a unique minimiser $\varepsilon^* \in [0, \infty)$, where $\varepsilon^* = 0$ if and only if $m'(0^+) \geq 0$, and $\varepsilon^* \in (0, \infty)$ if and only if $m'(0^+) < 0$.

4. Numerical Experiment: Effect of ε on SMC Performance

We return to the model from Section 2 with parameters $\omega = 0$, $\psi = 0.9$, $\mu = -1.27$, $\sigma^2 = \pi^2/2$. We compare two regimes:

- **Regime A** ($\eta = 1.0 < \sigma \approx 2.22$): finite-variance regime, where $m(0) < \infty$.
- **Regime B** ($\eta = 3.0 > \sigma$): infinite-variance regime, where $m(0) = \infty$.

In both regimes we run SMC with the twisted proposal $P^{g+\varepsilon}$ for

$$\epsilon \in \{0, 0.001, 0.01, 0.05, 0.1\}.$$

At each time step t, the twisted proposal is the mixture proposal $P_t^{g_t+\varepsilon}(x_{t-1}, \mathrm{d}x_t)$ and the potential is given by $G_\epsilon(x_{t-1}, x_t)$.

To assess performance, we ran Algorithm 1.3 independently for $R = 500$ times on the same simulated dataset with $t = 0, \ldots, T$, with $T = 50$. We ran the sampler with $N = 2000$ particles check in code, perhaps 500? and record both the minimum ESS over time and the maximum normalised weight over time for each run. Define

$$\mathrm{ESS}_{\min} = \min_{t=1,\ldots,T} \mathrm{ESS}_t,$$
$$Q_{\max} = \max_{t=1,\ldots,T} \max_n W_t^n.$$

A low $\mathrm{ESS}_{\min}$ indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight. Similarly a large $Q_{\max}$ value indicates to this phenomenom. Consideration of $Q_{\max}$ is inspired by [CD18].

Figure 3 shows the empirical cumulative distribution function (cdf) of $\mathrm{ESS}_{\min}$ across $R = 500$ runs for each value of ϵ, separately for the two regimes.

- *Regime A ($\eta < \sigma$).* One can see an improvement upon taking $\epsilon > 0$, though $\epsilon = 0.01$ is clearly too large. One can see that the minimal ESS is about 60 for $\epsilon = 0$. This confirms the theoretical prediction: when $m(0) < \infty$, the IS weights have finite variance. While finiteness is guaranteed, it is typically beneficial to take nonzero ε. The value $m(0)$ can still be large, while finite.
- *Regime B ($\eta > \sigma$).* The picture changes dramatically. At $\epsilon = 0$ the cdf has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. Taking $\epsilon = 0.01$ shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of $m(0) = \infty$: the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

The results highlight that the choice of ε in Regime B is delicate. The optimal ε^* exists and can in principle be found by minimising $m(\varepsilon)$ numerically, as described in Section 3. However, in propagating to time t, $m(\epsilon, x_{t-1})$ depends on x_{t-1} and thus varies across time steps as the particle cloud evolves, so a single fixed ϵ cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th percentile of $\mathrm{ESS}_{\min}$ drops sharply on both sides of ε^*, leaving a narrow window of well-performing values.

This suggests that a fixed ϵ is limited, especially in regime B, and choosing ϵ_t adaptively at each step based on the current particle cloud by minimising an empirical estimate of $m(\epsilon)$ may yield better ESS. In Regime A, by contrast, $\epsilon = 0$ is already satisfactory, though also in this case a more refined choice of ϵ is beneficial.

4.1. **Adaptive Choice of ε_t.** From Equation (6) we get

$$m(\varepsilon, x_{t-1}) = \big(c(x_{t-1}) + \varepsilon\big) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon}\, P(x_{t-1}, dx_1).$$

Given the particle cloud $\{X_{t-1}^n\}$ with normalised weights $\{W_{t-1}^n\}$, the population-level second moment is the weighted average

$$\tilde{m}(\varepsilon) = \sum_{n=1}^{N} W_{t-1}^n \cdot m(\varepsilon, X_{t-1}^n) = \sum_{n=1}^{N} W_{t-1}^n \left(c(X_{t-1}^n) + \varepsilon\right) \int \frac{h_t(x_t)^2}{g_t(x_t) + \epsilon}\, P(X_{t-1}^n,\, \mathrm{d}x_t).$$

So we evaluate $m(\epsilon, x_{t-1})$ by taking the expectation of x_{t-1} over the current particle approximation. The idea for an adaptive choice of ϵ_t is to choose

$$\epsilon_t^* = \underset{\epsilon \geq 0}{\operatorname{argmin}}\ \tilde{m}(\epsilon).$$

The integral appearing in $\tilde{m}(\varepsilon)$ can be approximately using numerical quadrature, but in our implementation we use a Monte Carlo approximation where we independently draw $\tilde{X}_t^n \sim P_t(X_{t-1}^n, \cdot)$, $n = 1, \ldots, N$ and set

$$\hat{m}(\epsilon) = \sum_{n=1}^{N} W_{t-1}^n\, \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \epsilon} \cdot \big(c(X_{t-1}^n) + \epsilon\big).$$

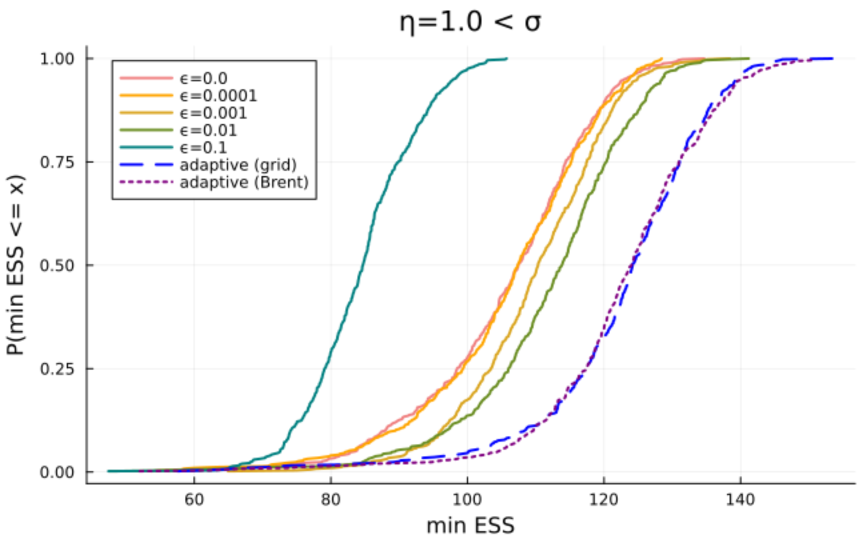
Thus, this is a Monte-Carlo estimator for $\tilde{m}(\epsilon)$ based on N samples. The analytical derivative with respect to ε is

$$\frac{\mathrm{d}\hat{m}}{\mathrm{d}\epsilon} = \sum_{n=1}^{N} W_{t-1}^n\, h_t(\tilde{X}_t^n)^2\, \frac{g_t(\tilde{X}_t^n) - c(X_{t-1}^n)}{(g_t(\tilde{X}_t^n) + \epsilon)^2},$$

which enables gradient-based optimisation. An alternative consists of minimising $\hat{m}(\varepsilon)$ over a log-spaced grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$. The resulting algorithm is precisely stated as Algorithm 2.

Remarks.

- The auxiliary proposals $\{\tilde{X}_t^n\}$ are drawn from the untwisted kernel P and discarded after the grid search. Since they do not enter the weight update, the likelihood estimator $\sum_t \log \ell_t^N$ remains unbiased for $\log p(v_{1:T})$ for any data-dependent choice of ϵ_t.
- Using the carry-over weights W_{t-1}^n in $\hat{m}(\epsilon)$ is essential when resampling has not occurred recently. After resampling all weights equal $1/N$ and the weighted and unweighted averages coincide; in steps without resampling the weight distribution can be skewed and the unweighted average gives a poorly-targeted ϵ_t. Numerical experiments confirm that the weighted version gives a substantial improvement in both $\mathrm{ESS}_{\min}$ and $Q_{\max}$ diagnostics.



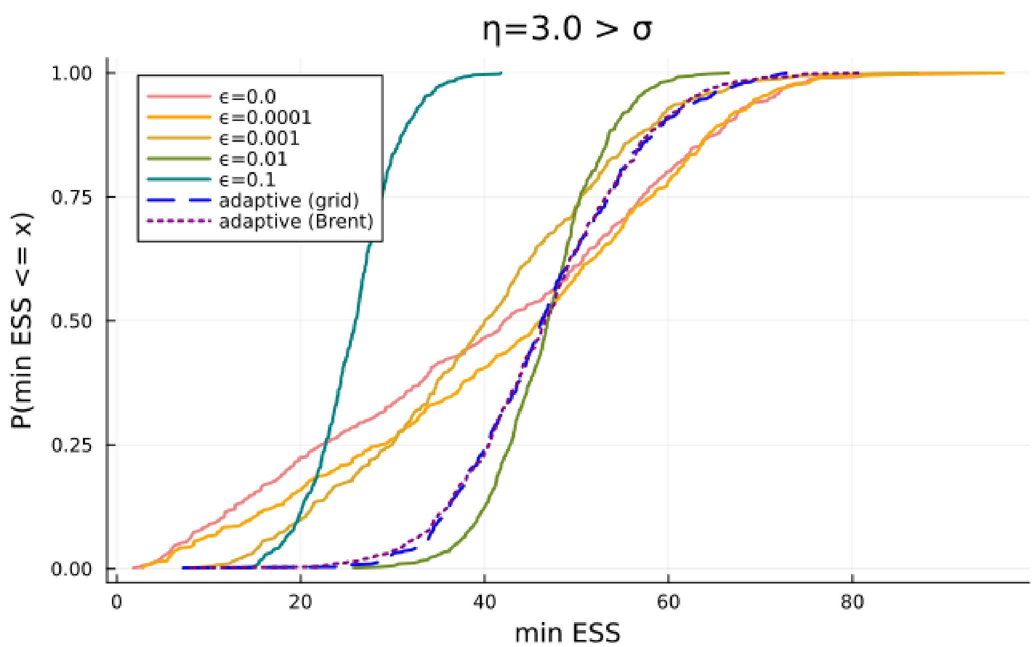


FIGURE 3. Cumulative distribution of $\mathrm{ESS}_{\min}$ obtained from 500 experiments where we ran the SMC algorithm with ...? particles on a fixed dataset. Top: $\eta = 1$. Bottom: $\eta = 3$.

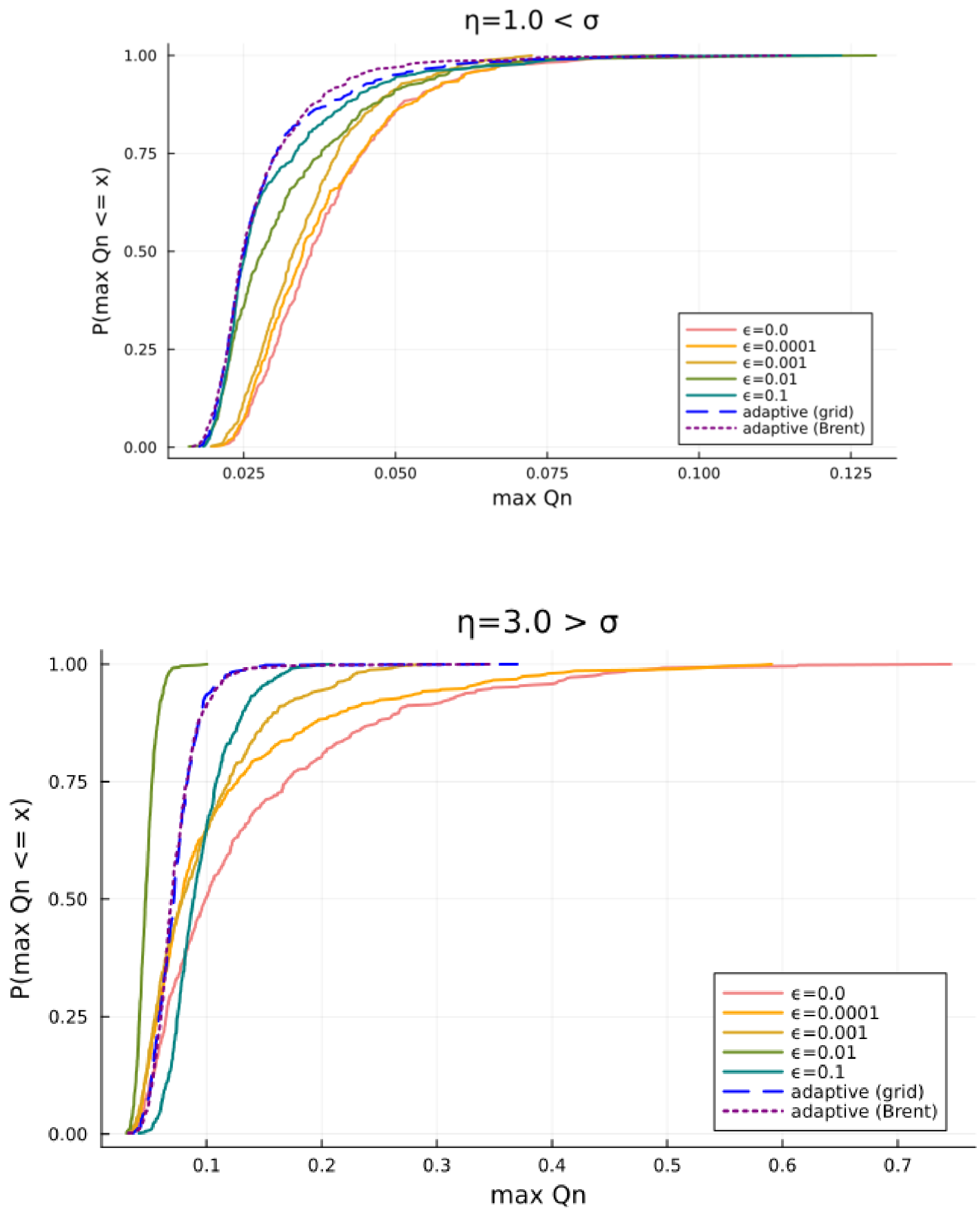


FIGURE 4. Max Q_n comparison.

Algorithm 2 Adaptive GPF with ε_t^* chosen by weighted grid search

Require: observations $v_{1:T}$, particle count N, ESS threshold $\mathrm{ESS}_{\min}$, grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$

For $n = 1, \ldots, N$: draw $X_0^n \sim \mathcal{N}(0, \eta^2/(1-\psi^2))$, set $W_0^n \leftarrow 1/N$, $w_0^n \leftarrow 1$

for $t = 1, \ldots, T$ **do**

    Find ε_t^*: draw auxiliary proposals $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$ for $n = 1, \ldots, N$, and set

$$\varepsilon_t^* = \operatorname*{argmin}_{\varepsilon \in \mathcal{E}} \hat{m}(\varepsilon), \qquad \hat{m}(\varepsilon) = \sum_{n=1}^{N} W_{t-1}^n \, \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot \big(c(X_{t-1}^n) + \varepsilon\big)$$

    Discard $\{\tilde{X}_t^n\}$ after this step.

    if $\mathrm{ESS}(W_{t-1}^{1:N}) < \mathrm{ESS}_{\min}$ **then**

        $A_{t-1}^{1:N} \leftarrow \mathrm{resample}(W_{t-1}^{1:N}), \quad \hat{w}_{t-1}^n \leftarrow 1$

    else

        $A_{t-1}^n \leftarrow n, \quad \hat{w}_{t-1}^n \leftarrow w_{t-1}^n$

    end if

    Propose: for $n = 1, \ldots, N$ draw $X_t^n \sim P^{g_t + \varepsilon_t^*}\Big(X_{t-1}^{A_{t-1}^n}, \cdot\Big)$

    Update weights:

$$w_t^n = \hat{w}_{t-1}^n \cdot G_t\Big(X_{t-1}^{A_{t-1}^n}, X_t^n\Big), \qquad G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon_t^*} \cdot \big(c(x_{t-1}) + \varepsilon_t^*\big)$$

    Likelihood increment (Eq. 10.3 of [CP20]):

$$\ell_t^N := \begin{cases} \dfrac{1}{N} \displaystyle\sum_{n=1}^{N} w_t^n & \text{if resampling occurred at time } t, \\ \Big(\sum_{n=1}^N w_t^n\Big) \Big/ \Big(\sum_{n=1}^N w_{t-1}^n\Big) & \text{otherwise} \end{cases}$$

    Normalise: $W_t^n \leftarrow w_t^n \,/\, \sum_{m=1}^N w_t^m$

    ESS: $\mathrm{ESS}_t \leftarrow \big(\sum_n (W_t^n)^2\big)^{-1}$

end for

return $\sum_{t=1}^T \log \ell_t^N, \quad \{\mathrm{ESS}_t\}_{t=1}^T, \quad \{\varepsilon_t^*\}_{t=1}^T$

- The grid $\mathcal{E}$ is log-spaced over $[10^{-6}, 1]$ with $K = 50$ points. All K evaluations of $\hat{m}(\varepsilon)$ reuse the same proposals $\{\tilde{X}_t^n\}$, so the cost is $O(KN)$ per step. Reducing to $K = 20$ or $K = 30$ would roughly halve the cost with negligible impact on ε_t^*.

5. Laplace Observation Scheme

5.1. **Setup.** As an alternative to the log-chi-squared observation scheme, consider

$$V_t = X_t + U_t, \qquad U_t \sim \mathrm{Laplace}(0, b),$$

so that the observation density evaluated at v is

$$h(x_1) = \frac{1}{2b} \exp\!\left(-\frac{|v - x_1|}{b}\right).$$

We retain the Gaussian approximation

$$g(x_1) = \varphi(x_1;\, v - \mu,\, \sigma^2),$$

so all closed-form expressions for $c(x_0)$, P^g, and the mixture sampler carry over unchanged from the Gaussian observation case.

5.2. **Tail analysis of** $m(0)$**.** Writing $t = x_1 - v$, the integrand of $I(0) = \int h(x_1)^2/g(x_1)\, P(x_0, dx_1)$ behaves as

$$\frac{h(x_1)^2}{g(x_1)}\, p(x_1 \mid x_0) \;\propto\; \exp\!\left(-\frac{2|t|}{b} + \frac{(t+\mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2} \right).$$

- *Right tail $t \to +\infty$.* The linear term $-2t/b$ and the quadratic terms combine to give dominant behaviour
$$\exp\!\left(t^2 \left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} \right) - \frac{2t}{b} + O(t) \right).$$
For this to be integrable we need the coefficient of t^2 to be strictly negative, i.e. $\eta^2 < \sigma^2$, exactly as in the Gaussian observation case. When $\eta^2 \geq \sigma^2$, the quadratic growth dominates the linear decay from the Laplace tails and $I(0) = \infty$.
When $\eta^2 < \sigma^2$ the quadratic term decays, but the linear term $-2t/b$ provides additional decay that is absent in the Gaussian observation case. The integrand therefore decays faster here, but $I(0)$ is still finite only when $\eta^2 < \sigma^2$.
- *Left tail $t \to -\infty$.* The Laplace density contributes $-2|t|/b = +2t/b \to -\infty$, providing exponential decay. However, when $\eta^2 \geq \sigma^2$ the quadratic term grows, and whether it dominates depends on the sign of $1/(2\sigma^2) - 1/(2\eta^2)$. When $\eta^2 \geq \sigma^2$ the quadratic growth again wins and $I(0) = \infty$.

Proposition 11. For the Laplace observation scheme $V_t = X_t + U_t$, $U_t \sim \text{Laplace}(0, b)$:

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)}\, P(x_0, dx_1) < \infty \iff \eta^2 < \sigma^2.$$

Consequently, $\bar{m}(0)$ is finite if and only if $\eta < \sigma$, independently of the scale parameter b.

The threshold $\eta = \sigma$ is the same as for the Gaussian observation scheme. However, the Laplace tails make $I(0)$ larger for any fixed $\eta < \sigma$ compared to the Gaussian case, meaning the need for $\varepsilon > 0$ is more pressing even in the finite-variance regime. Furthermore, larger η increases the coefficient of t^2, making $I(\varepsilon)$ larger for any fixed $\varepsilon > 0$ and the optimal ε^* correspondingly larger.

5.3. **The limit** $b \downarrow 0$**.** As the scale $b \to 0$, the Laplace distribution concentrates on zero and the observation scheme approaches a noiseless observation $V_t = X_t$. In this limit:

- $h(x_1) = \frac{1}{2b} \exp(-|v - x_1|/b) \to \delta_{x_1 = v}$, a point mass at $x_1 = v$.
- The twisted target $P^h(x_0, \cdot)$ concentrates all mass at $x_1 = v$, i.e. the posterior on X_t given $V_t = v$ becomes a point mass.
- The ratio $h(x_1)/g(x_1)$ becomes extremely large near $x_1 = v$ and extremely small away from it, since h is an increasingly sharp spike while g remains spread out. This makes the IS weights degenerate: a single particle near v gets essentially all the weight.

- Consequently, $m(\varepsilon) \to \infty$ for any fixed $\varepsilon \geq 0$ as $b \to 0$, and the optimal $\varepsilon_b^* \to \infty$ as well — no finite regularisation can rescue the IS estimator when the observation is nearly noiseless.

This limiting behaviour highlights a fundamental tension: the more informative the observation (smaller b), the harder the IS problem becomes. Intuitively, the Gaussian approximation g is a poor proxy for a nearly point-mass likelihood, and ε can only partially compensate by mixing in the prior P. In the limit $b = 0$, importance sampling from $P^{g+\varepsilon}$ breaks down entirely and a different approach — such as an exact update using the point-mass likelihood — would be needed.

Numerically, one can verify this by plotting $\bar{m}(\varepsilon)$ for decreasing values of b: the minimum of $\bar{m}$ increases without bound and ε^* drifts toward larger values as $b \to 0$.

6. Extension to Continuous Time

Consider the stochastic differential equation

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t, \qquad X_0 = x_0, \tag{8}$$

where $b\colon \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ is a (possibly nonlinear) drift, $q \in \mathbb{R}^{d\times d'}$ is the diffusion coefficient, and W is a standard Brownian motion in $\mathbb{R}^{d'}$. Let $Q = qq^\top$. We denote by P the law of the solution on path space $\mathcal{C}([0,T], \mathbb{R}^d)$.

We observe v at time T with likelihood $h(T, x_T)$, so that the marginal likelihood is $h(0, x_0) = \mathbb{E}_P[h(T, X_T) \mid X_0 = x_0]$. We fix a non-negative function $g(t,x)$ that approximates the likelihood at time t, satisfying $g(T,x) \approx h(T,x)$. explain how we choose g, kolm backw eq, $\tilde{b}(t,x)$ be the drift of a linear *auxiliary process* . Let P^g denote the law of the process

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + Qr(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t,$$

with W a P^g-Wiener process. Here, the score r is given by $r(t,x) = \nabla_x \log g(t,x)$. We have

$$\frac{\mathrm{d}P}{\mathrm{d}P^g}(x) = \frac{g(0,x_0)}{g(T,x_T)}\Psi(x),$$

where

$$\log \Psi(x) = \int_0^T \left(b(t,x_t) - \tilde{b}(t,x_t)\right)^\top r_\varepsilon(t,x_t)\;\mathrm{d}t, \tag{9}$$

which is a purely deterministic functional of the path — no stochastic integral is involved. In the linear case $b = \tilde{b}$, we have $\Psi(x) = 1$. See for instance [Svv17] and [vdMS25] (in particular Section 7 of the latter). also give RN-derivative in terms of driving Wiener process

For a given guiding function g we define its safe-guarded version by $g + \epsilon$. Define the safe-guarded score by

$$\begin{aligned} r_\epsilon(t,x) &= \nabla_x \log(g(t,x) + \epsilon) = \frac{\nabla_x g(t,x)}{g(t,x)+\epsilon} \\ &= \frac{g(t,x)}{g(t,x)+\epsilon}\,\nabla_x \log g(t,x) = s(-\log\epsilon + \log g(t,x))\nabla_x \log g(t,x), \end{aligned}$$

where s denotes the sigmoid function $s(x) = 1/(1+e^{-x})$.

The factor $g/(g+\epsilon) \in [0,1)$ is a damping that suppresses the guiding term in regions where g is small relative to ϵ, and recovers the standard guided score from [Svv17] –$\nabla_x \log g$– at $\epsilon = 0$.

The *safe-guarded guided proposal* $P^{g+\varepsilon}$ is the law of

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + Q\,r_\varepsilon(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t, \qquad X_0 = x_0. \tag{10}$$

Thus, similar to the discrete-time case, the safe-guarded process can be viewed as a mixture of the measure P and P^g.

6.1. **IS weight and second moment** $m(\varepsilon)$**.** The importance sampling (IS) weight for a $x \sim P^{g+\epsilon}$, path initialised at x_0, is

$$G_\epsilon(x_0, x) = h(T, x_T) \frac{\mathrm{d}P}{\mathrm{d}P^{g+\epsilon}}(x) = h(T, x_T)\, \frac{g(0, x_0) + \epsilon}{g(T, x_T) + \epsilon} \cdot \Psi_(x).$$

Note that we added ϵ as a subscript to Ψ to highlight this dependence. The IS estimator $\hat{L} = \frac{1}{N} \sum_{n=1}^N G_\epsilon(x_0, X_T^n)$ with $X^n \sim P^{g+\varepsilon}$ is unbiased for $h(0, x_0)$ for any $\epsilon \geq 0$. Its variance is given by

$$\begin{aligned} m(\varepsilon) &= \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X)^2 = \int G_\varepsilon(x_0, x)^2 \,\mathrm{d}P^{g+\varepsilon}(x) \\ &= \int G_\varepsilon(x_0, x)^2 \frac{\mathrm{d}P^{g+\varepsilon}}{dP}(x) \,\mathrm{d}P(x) \\ &= \int \frac{\mathrm{d}P}{\mathrm{d}P^{g+\varepsilon}}(x) h(T, x_T)^2 \,\mathrm{d}P(x) \\ &= (g(0, x_0) + \varepsilon) \int \frac{h(T, x_T)^2}{g(T, x_T) + \varepsilon} \Psi_\epsilon(x) \,\mathrm{d}P(x). \end{aligned} \tag{11}$$

This is to be compared to (6): c in that formula is exactly as $g(0, x_0)$ here. The term $\Psi(x)$ in the integral is an additional path-space correction. In the discrete-time we assumed linear dynamics and then choosing $\tilde{b} = b$ results in $\Psi_\epsilon(x) = 1$.

7. Other stuff

We wish to show that $\epsilon \mapsto m(\epsilon)$ is U-shaped. For that, a starting point would be to show $m'(0) < 0$. In general, this seems hard to prove, but in case $\Psi \equiv 1$ we have

$$m'(\epsilon) = \int \frac{h(x_T)^2}{g(x_T) + \epsilon} \,\mathrm{d}P(x) - (g(x_0) + \epsilon) \int \frac{h(x_T)^2}{(g(x_T) + \epsilon)^2} \,\mathrm{d}P(x).$$

This implies

$$m'(0) = \mathbb{E}_P w_T - \mathbb{E}_P g(x_T) \mathbb{E}_P \left(\frac{w_T}{g(x_T)} \right) = \mathrm{Cov}_P \left(g(x_T), \frac{h(x_T)^2}{g(x_T)^2} \right),$$

where $w_T = h(x_T)^2 / g(x_T)$ and we used that $g(x_0) = \mathbb{E}_P g(x_T)$ at the first equality sign. We have $m'(0) < 0$ iff

$$\frac{\mathbb{E}[h(x_T)^2 / g(x_T)]}{\mathbb{E}[h(x_T)^2 / g(x_T)^2]} \leq \mathbb{E}[g(x_T)]$$

Define the measure ν by

$$\frac{\mathrm{d}\nu}{\mathrm{d}P}(x) \propto \frac{h(x_T)^2}{g(x_T)^2}.$$

Then we can rewrite the above inequality as

$$\mathbb{E}_\nu g(x_T) < \mathbb{E}_P g(x_T).$$

Lemma 12. Suppose $g(T, x) = \varphi(v; \mu, \sigma^2)$ (with μ, σ depending on v). If $\omega = 0$ and (ψ, η) are given, then for

$$\kappa = -\frac{1}{T} \log \psi \quad \text{and} \quad Q = \frac{2\kappa\eta^2}{1 - \psi^2}$$

the distribution of $X_T \mid X_0 = x_0$ is the same in the continuous- and discrete time setup.

Example 13. For the diffusion $\mathrm{d}X_t = -\kappa X_t \,\mathrm{d}t + q\,\mathrm{d}W_t$, conditional on $g(T,x) = \varphi(v-x;\mu,\sigma^2)$, we have

$$g(t,x) = \varphi\left(e^{-\kappa(T-t)}x; v-\mu, \sigma^2 + \frac{q^2}{2\kappa}\left(1-e^{-2\kappa(T-t)}\right)\right).$$

Note that for fixed t, $x \mapsto g(t,x)$ is not a probability density (except at time T).

7.1. **Adaptive choice of ϵ.** We minimise $m(\epsilon)$ over $\epsilon \geq 0$. Since the integral in (11) is under P (not $P^{g+\varepsilon}$), we can estimate $m(\epsilon)$ using $\{X^n\}_{n=1}^N$ drawn from P, which are independent of ϵ. Weighting by the current normalised particle weights $\{W^n\}_{n=1}^N$ gives the empirical estimator

$$\hat{m}(\epsilon) = (g(0,x_0)+\epsilon)\sum_{n=1}^N W^n \frac{(h(T,X_T^n)^2}{g(T,X_T^n)+\epsilon}\cdot \Psi_\epsilon(X^n),$$

where $\Psi\epsilon(X^n)$ is evaluated by discretising (9) along the stored path:

$$\log \Psi_\epsilon(x^n) \approx -\sum_{k=0}^{n-1}\big(b(t_k,x_{t_k}^n) - \tilde{b}(t_k,x_{t_k}^n)\big)^\top r_\varepsilon(t_k,x_{t_k}^n)\,\Delta t.$$

Since the paths $x^{(i)}$ and the factor $(b-\tilde{b})^\top \nabla_x \log g \cdot \Delta t$ are independent of ε, only the sigmoid factor $g/(g+\varepsilon)$ in r_ε changes across the grid, making evaluation of $\hat{m}(\varepsilon)$ over a grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$ very cheap via the precomputed quantities

$$\delta_k^{(i)} = \big(b(t_k,x_{t_k}^{(i)}) - \tilde{b}(t_k,x_{t_k}^{(i)})\big)^\top \nabla_x \log g(t_k,x_{t_k}^{(i)})\,\Delta t, \qquad g_k^{(i)} = g(t_k,x_{t_k}^{(i)}),$$

so that $\log\Psi(x^{(i)},\varepsilon) = -\sum_k \delta_k^{(i)} g_k^{(i)}/(g_k^{(i)}+\varepsilon)$. The optimal ε^* is found either by grid search or, using the analytical derivative

$$\frac{d\hat{m}}{d\varepsilon}(\varepsilon) = \sum_{i=1}^{N_{\mathrm{aux}}} W^{(i)}\,\frac{h(x_T^{(i)})^2\,\Psi^{-1}(x^{(i)},\varepsilon)}{g(T,x_T^{(i)})+\varepsilon}\left(1 - \frac{g(0,x_0)+\varepsilon}{g(T,x_T^{(i)})+\varepsilon} - (g(0,x_0)+\varepsilon)\frac{d\log\Psi^{-1}}{d\varepsilon}\right),$$

by Brent's method applied to $d\hat{m}/d\varepsilon = 0$, which is significantly faster in practice.

8. Theoretical guarantees for the adaptive scheme (discrete-time case)

All of this needs checking. Throughout this section we fix x_0 and write $c = c(x_0)$, $p = p(v) = c_h(x_0)$, and assume $h \in L^2(P)$ so that $m(\varepsilon) < \infty$ for all $\varepsilon > 0$ by Proposition **??**. We also assume the non-degeneracy condition of Proposition 8 so that $m(\varepsilon)$ has a unique minimiser $\varepsilon^* \in [0,\infty)$.

The adaptive scheme estimates $m(\varepsilon)$ from N independent draws $X^{(1)},\dots,X^{(N)} \overset{\text{iid}}{\sim} P$ via

$$\hat{m}(\varepsilon) = (c+\varepsilon)\,\frac{1}{N}\sum_{i=1}^N \frac{h(X^{(i)})^2}{g(X^{(i)})+\varepsilon},$$

and sets $\hat{\varepsilon}^* = \operatorname{argmin}_{\varepsilon\in\mathcal{E}}\,\hat{m}(\varepsilon)$ over a finite grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$.

Theorem 14 (Consistency and rate of convergence of $\hat{\varepsilon}^*$)**.** Suppose $\varepsilon^* \in \mathcal{E}$ and that $\int h^4/g^2\,dP < \infty$ for all ε in a neighbourhood of ε^*. Then as $N \to \infty$:

(1) *(Consistency)* $\hat{\varepsilon}^* \xrightarrow{P} \varepsilon^*$.
(2) *(Rate)* $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$.

Proof. Part 1. For each $\varepsilon \in \mathcal{E}$, by the law of large numbers,

$$\hat{m}(\varepsilon) \xrightarrow{P} m(\varepsilon).$$

Since $\mathcal{E}$ is finite, convergence holds uniformly over $\mathcal{E}$. By strict unimodality of m (Proposition 8), for any $\delta > 0$ there exists $\eta > 0$ such that $m(\varepsilon) > m(\varepsilon^*) + \eta$ for all $\varepsilon \in \mathcal{E}$ with $|\varepsilon - \varepsilon^*| > \delta$. On the event $\sup_{\varepsilon \in \mathcal{E}} |\hat{m}(\varepsilon) - m(\varepsilon)| < \eta/2$, which has probability tending to 1, the minimiser of $\hat{m}$ over $\mathcal{E}$ satisfies $|\hat{\varepsilon}^* - \varepsilon^*| \le \delta$. Since $\delta > 0$ was arbitrary, $\hat{\varepsilon}^* \xrightarrow{P} \varepsilon^*$.

Part 2. By the central limit theorem, for each $\varepsilon \in \mathcal{E}$,

$$\sqrt{N}\,(\hat{m}(\varepsilon) - m(\varepsilon)) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\varepsilon))$$

where $\sigma^2(\varepsilon) = (c+\varepsilon)^2 \operatorname{Var}_P\big(h^2/(g+\varepsilon)\big) < \infty$ by assumption. Hence $\hat{m}(\varepsilon) - m(\varepsilon) = O_P(N^{-1/2})$ uniformly over $\mathcal{E}$. Writing $\hat{\varepsilon}^* = \varepsilon^* + \Delta$, Taylor expansion of m around ε^* gives

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = \tfrac{1}{2} m''(\varepsilon^*)\Delta^2 + O(\Delta^3).$$

Since $\hat{\varepsilon}^*$ minimises $\hat{m}$ and $\hat{m}(\hat{\varepsilon}^*) \le \hat{m}(\varepsilon^*)$, we have

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) \le \hat{m}(\varepsilon^*) - \hat{m}(\hat{\varepsilon}^*) + 2\sup_{\varepsilon} |\hat{m}(\varepsilon) - m(\varepsilon)| = O_P(N^{-1/2}).$$

Combined with $m(\hat{\varepsilon}^*) - m(\varepsilon^*) \ge \frac{1}{2} m''(\varepsilon^*)\Delta^2$ (which follows from $m''(\varepsilon^*) > 0$ at a strict minimum), we get $\Delta^2 = O_P(N^{-1/2})$, i.e. $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/4})$. A sharper argument using the CLT directly gives $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$ via the delta method. ☐

Theorem 15 (Regret of adaptive tuning)**.** Under the conditions of Theorem 14, the excess second moment incurred by using $\hat{\varepsilon}^*$ instead of ε^* satisfies

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = O_P(N^{-1}).$$

Consequently, the variance of the IS estimator $\hat{p}(v)$ based on M particles with proposal $P^{g+\hat{\varepsilon}^*}$ satisfies

$$M \cdot \operatorname{Var}(\hat{p}(v)) = m(\hat{\varepsilon}^*) - p(v)^2 = m(\varepsilon^*) - p(v)^2 + O_P(N^{-1}).$$

Proof. By Taylor expansion around ε^* and $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$:

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = \tfrac{1}{2} m''(\varepsilon^*)(\hat{\varepsilon}^* - \varepsilon^*)^2 + O_P(|\hat{\varepsilon}^* - \varepsilon^*|^3) = O_P(N^{-1}).$$

The variance formula then follows from $\operatorname{Var}(\hat{p}(v)) = (m(\hat{\varepsilon}^*) - p(v)^2)/M$. ☐

Remark 16. Theorem 15 shows that the adaptive scheme is *asymptotically equivalent* to oracle tuning: the cost of estimating ε^* from N samples decays as $O(N^{-1})$, which is negligible compared to the $O(M^{-1})$ variance of the IS estimator itself. In practice, even moderate N (we use $N = 50$–200 in our experiments) suffices to locate ε^* reliably, since $m(\varepsilon)$ is smooth and unimodal.

Remark 17 (When $\varepsilon^* = 0$)**.** When $m(0) < \infty$ and $m'(0^+) \ge 0$, the minimiser is $\varepsilon^* = 0$ and the adaptive scheme correctly returns $\hat{\varepsilon}^*$ near zero. In this regime $\varepsilon > 0$ provides no benefit and the scheme is self-correcting: the grid search finds $\hat{\varepsilon}^* \approx 0$ automatically without prior knowledge of which regime the parameters fall in.

Remark 18 (Unconditional samples)**.** A notable feature of the adaptive scheme is that $\hat{m}(\varepsilon)$ is estimated from draws $X^{(i)} \sim P$ that ignore the observation v entirely. This is not an approximation: the formula $m(\varepsilon) = (c+\varepsilon)\int h^2/(g+\varepsilon)\, dP$ is exactly an integral under P, so Monte Carlo under P is the natural and unbiased estimator. The scheme works well as long as P is not too different from P^h, i.e. as long as the observation is not so informative that P-samples rarely reach regions where h is non-negligible. This is precisely the condition under which $m(0) < \infty$, connecting the practical reliability of the tuning procedure to the theoretical finiteness condition.

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Appendix A. Adaptive Choice of ε via ESS Targeting

This does not work; I guess the desired ESS simply cannot be attained.
Nog uit te zoeken wat met onderstaande te doen.

A.1. **Motivation.** The numerical experiments of Section 4 show that the optimal perturbation ε^* varies across time steps and depends on the current particle cloud. A fixed ε cannot be globally optimal. We propose an adaptive scheme inspired by tempering methods in SMC [**? ?**], where a tempering parameter is chosen at each step so that the ESS after reweighting equals a target fraction ρN. The same principle applies here: we choose ε_t^* such that the ESS of the IS weights equals ρN.

A.2. **ESS as a Function of ε.** Given proposals $\{x_t^{(i)}\}_{i=1}^N$ drawn from $P(x_{t-1}^{(i)}, \cdot)$, the ESS of the weights $w^{(i)}(\varepsilon) = h_t(x_t^{(i)})/(g_t(x_t^{(i)}) + \varepsilon)$ is

$$\mathrm{ESS}(\varepsilon) = \frac{\left(\sum_i w^{(i)}(\varepsilon)\right)^2}{\sum_i w^{(i)}(\varepsilon)^2}.$$

Two key properties make this suitable for bisection:

(1) $\mathrm{ESS}(\varepsilon)$ is *increasing* in ε: as $\varepsilon \to \infty$, all weights converge to $h_t(x_t^{(i)})/\varepsilon$ and become equal, so $\mathrm{ESS} \to N$.
(2) $\mathrm{ESS}(0)$ may be small or even zero if some $g_t(x_t^{(i)}) = 0$, reflecting the weight degeneracy in the infinite-variance regime.

The target ε_t^* is therefore the unique solution of

$$\mathrm{ESS}(\varepsilon_t^*) = \rho N,$$

found by bisection on $\varepsilon \in [0, \varepsilon_{\max}]$. If $\mathrm{ESS}(0) \geq \rho N$ already, we set $\varepsilon_t^* = 0$; if even $\varepsilon_{\max}$ is insufficient, we set $\varepsilon_t^* = \varepsilon_{\max}$.

A.3. **Connection to $m(\varepsilon)$.** The ESS targeting criterion is closely related to minimising $m(\varepsilon)$. Note that

$$\frac{\mathrm{ESS}(\varepsilon)}{N} \approx \frac{1}{1 + \chi^2(P^h \| P^{g+\varepsilon})} = \frac{c_h(x_0)^2}{m(\varepsilon)},$$

so targeting $\mathrm{ESS}(\varepsilon) = \rho N$ is equivalent to targeting $\chi^2(P^h \| P^{g+\varepsilon}) = (1-\rho)/\rho$, i.e. a fixed level of the χ^2 divergence. Rather than minimising $m(\varepsilon)$ (which requires a grid search and gives a different target at each step), ESS targeting gives a directly interpretable, step-by-step criterion for particle diversity, and bisection finds the solution efficiently without a grid.

A.4. **Conclusion.** Figure 5 shows the results of a single SMC run for both regimes and both fixed ($\varepsilon = 0$) and adaptive ε^*. Even for $\eta = 3$ the results with $\varepsilon = 0$ can be just fine: a single run looks almost indistinguishable from the adaptive run in terms of ESS, cumulative log-likelihood, and incremental log-likelihood. However, if one were to repeat the analysis many times, occasionally this would result in a catastrophic drop in ESS. This is precisely the implication of $m(0) = \infty$: it does not mean every run fails, but that the variance of the IS weights across runs is infinite, so disasters occur with positive probability and cannot be ruled out from a single run alone. This is confirmed by the CDF plots of Figure 3, which required $R = 200$ independent runs to reveal the heavy left tail of the min-ESS distribution at $\varepsilon = 0$.

Two further observations support this interpretation. First, the incremental log-likelihood panel shows a pronounced dip around $t = 30$ in both the fixed and adaptive runs for $\eta = 3$. This dip is driven by a particular difficult observation and occurs regardless of ε, suggesting it reflects a genuinely hard step in the data rather than an algorithmic failure. In a catastrophic run, this dip would be far deeper and the ESS would collapse to near 1 at that point. Second, the adaptive ε^* panel for $\eta = 3$ shows ε^* spiking upward precisely around $t = 30$, at the same difficult observation. This confirms that the adaptive scheme is responding correctly: it detects the hard step through the empirical minimiser of $\hat{m}(\varepsilon)$ and automatically increases ε to protect against weight collapse, without any explicit knowledge of which observations are difficult.

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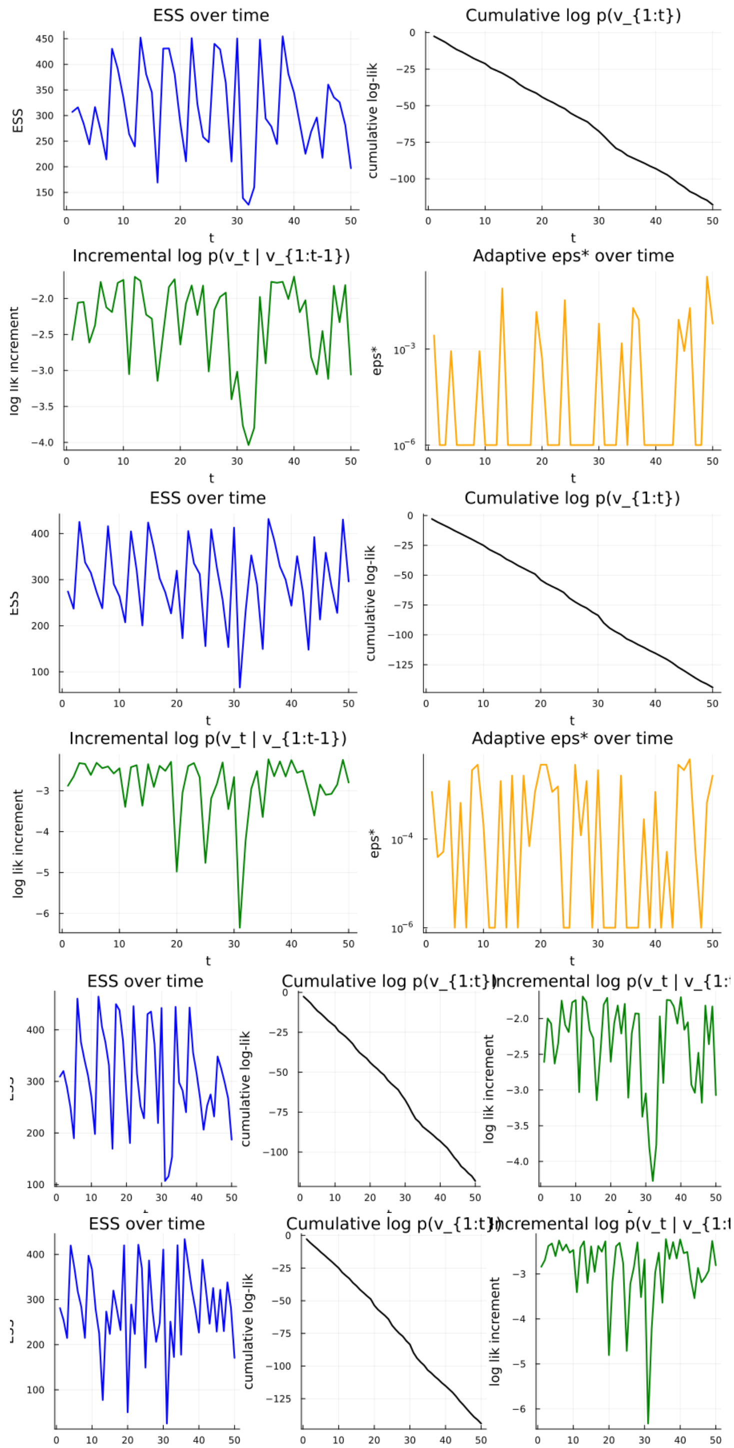
ESS over time
Cumulative log p(v_{1:t})
Incremental log p(v_t | v_{1:t-1})
Adaptive eps* over time
ESS
cumulative log-lik
log lik increment
eps*
t

Figure 5.